

### 第三章 第一节 不定积分

#### 一、选择题

1. 答案: A

解析: 由  $F'(x) = G'(x)$ , 可得  $F'(x) - G'(x) = 0$ , 即  $[F(x) - G(x)]' = 0$ , 导数恒为零的函数为常数函数, 因此  $F(x) - G(x) = C$ , 故选 A.

2. 答案: B

解析:

B 选项: 根据不定积分的定义, 若  $F'(x) = f(x)$ , 则  $\int f(x)dx = F(x) + C$ , 正确, 故选 B.

A 选项:  $\int F'(x)dx = f(x) + C$ , 错误, 应为  $\int F'(x)dx = F(x) + C$ .

C 选项:  $\int F(x)dx = f(x) + C$ , 错误, 应为  $\int f(x)dx = F(x) + C$ .

D 选项:  $\int f'(x)dx = F(x) + C$ , 错误, 应为  $\int f'(x)dx = f(x) + C$ .

3. 答案: C

解析: 两边同时求导  $\left(\int f(x)dx\right)' = (x \ln x + C)'$ , 得  $f(x) = \ln x + x \cdot \frac{1}{x}$ , 即  $f(x) = \ln x + 1$ , 所以  $f'(x) = \frac{1}{x}$ , 故选 C.

4. 答案: A

解析: 根据原函数与不定积分的定义, 若  $\sin x^2$  是  $f(x)$  的一个原函数, 则  $\int f(x)dx = \sin x^2 + C$ , 故选 A.

5. 答案: A

解析:  $\int f(x)dx = \int (x + \sqrt{x})dx = \int xdx + \int x^{\frac{1}{2}}dx = \frac{x^2}{2} + \frac{2}{3}x^{\frac{3}{2}} + C$ , 故选 A.

6. 答案: D

解析:

D 选项: 根据不定积分的求导法则,  $\frac{d}{dx} \int f(x) dx = \left( \int f(x) dx \right)' = f(x)$ , 正确, 故选 D.

A 选项: 错误, 应为  $d \int f(x) dx = f(x) dx$ , 原式子缺少积分常数  $dx$ .

B 选项: 错误, 应为  $\int f'(x) dx = f(x) + C$ , 原式子缺少积分常数  $C$ .

C 选项: 错误, 应为  $\int df(x) = f(x) + C$ , 式子缺少积分常数  $C$ .

7. 答案: A

解析: 原式  $= \int \frac{x^2+1}{x} dx = \int \left( x + \frac{1}{x} \right) dx = \int x dx + \int \frac{1}{x} dx = \frac{x^2}{2} + \ln|x| + C$ , 故选 A.

8. 答案: A

解析: 原式  $= \int \tan^2 x dx = \int (\sec^2 x - 1) dx = \int \sec^2 x dx - \int 1 dx = \tan x - x + C$ , 故选 A.

9. 答案: A

解析: 原式

$$= \frac{1}{2} \int \frac{1}{x^2+2x+3} (x^2+2x+3)' dx = \frac{1}{2} \int \frac{1}{x^2+2x+3} d(x^2+2x+3) = \frac{1}{2} \ln(x^2+2x+3) + C, \text{ 故选 A.}$$

10. 答案: A

解析: 原式  $= \int x f(x^2) dx = \frac{1}{2} \int f(x^2) (x^2)' dx = \frac{1}{2} \int f(x^2) dx^2 = \frac{1}{2} F(x^2) + C$ , 故选 A.

11. 答案: A

解析: 由  $\int f(x) dx = x^3 + C$  两边求导得  $\left( \int f(x) dx \right)' = (x^3 + C)'$ , 即  $f(x) = 3x^2$ , 则

$$\int f(\cos x) \sin x dx = - \int f(\cos x) (\cos x)' dx = - \int f(\cos x) d \cos x = - \cos^3 x + C, \text{ 故选 A.}$$

12. 答案: A

解析：原式  $= \int \frac{1}{1+e^x} (1+e^x)' dx = \int \frac{1}{1+e^x} d(1+e^x) = \ln(1+e^x) + C$ ，故选 A.

13. 答案: A

解析：原式  $= x \arcsin x - \int \frac{1}{\sqrt{1-x^2}} \cdot x dx = x \arcsin x + \frac{1}{2} \int \frac{1}{\sqrt{1-x^2}} d(1-x^2)$   
 $= x \arcsin x + \frac{1}{2} \cdot 2\sqrt{1-x^2} + C = x \arcsin x + \sqrt{1-x^2} + C$ ，故选 A.

14. 答案: D

解析：若为同一函数的原函数，则求导后应为同一个函数.

D 选项：  $(-\cos^2 x)' = -2 \cos x \cdot (-\sin x) = \sin 2x$ ，  $(\sin^2 x)' = 2 \sin x \cos x = \sin 2x$ ，导数相等，正确，故选 D.

A 选项：  $(\sqrt{x+1})' = \frac{1}{2\sqrt{x+1}}$ ，  $(2\sqrt{x+1})' = 2 \cdot \frac{1}{2\sqrt{x+1}} = \frac{1}{\sqrt{x+1}}$ ，导数不相等，错误.

B 选项：  $(\ln x^2)' = \frac{2}{x}$ ，  $(\ln 2x)' = \frac{1}{2x} \cdot 2 = \frac{1}{x}$ ，导数不相等，错误.

C 选项：  $(e^{3x})' = 3e^{3x}$ ，  $(3e^{3x})' = 9e^{3x}$ ，导数不相等，错误.

15. 答案: A

解析：原式  $= \int \frac{1}{\ln(\ln x)} \cdot \frac{1}{x \ln x} dx = \int \frac{1}{\ln(\ln x)} [\ln(\ln x)]' dx = \int \frac{1}{\ln(\ln x)} d \ln(\ln x) = \ln |\ln(\ln x)| + C$ ，故选 A.

16. 答案: B

解析：由题可得，  $\lim_{h \rightarrow 0} \frac{F(1+h) - F(1-h)}{h} = \frac{(1+h) - (1-h)}{h} F'(1) = 2F'(1)$ ，因为

$F(x) = x \sin x + 2 \cos x$ ，所以  $F'(x) = \sin x + x \cos x - 2 \sin x = x \cos x - \sin x$ ，

$F'(1) = \cos 1 - \sin 1$ ，得原式  $= 2(\cos 1 - \sin 1)$ ，故选 B.

17. 答案: B

解析: 先求极限  $f(x) = \lim_{n \rightarrow \infty} n \ln \left( 1 + \frac{x}{n} \right) = \lim_{n \rightarrow \infty} n \cdot \frac{x}{n} = x$ , 故  $\int f(x) dx = \int x dx = \frac{1}{2} x^2 + C$ , 故选 B.

18. 答案: C

解析: 由  $f'(x) = 2e^x$ , 得  $f(x) = \int 2e^x dx = 2e^x + C$ , 代入  $f(0) = 1$ ,  $2e^0 + C = 1$ , 得  $C = -1$ , 即

$f(x) = 2e^x - 1$ , 则  $\int f(x) dx = \int (2e^x - 1) dx = 2e^x - x + C$ , 故选 C.

## 二、填空题

1. 答案:  $2x \sin x + x^2 \cos x + C$

解析: 由题可得,  $f'(x) = 2x \sin x + x^2 \cos x$ , 根据不定积分定义,  $\int f''(x) dx = f'(x) + C$

$= 2x \sin x + x^2 \cos x + C$ .

2. 答案:  $\frac{1}{2}$

解析: 两边求导  $f'(x) = \left( \int e^{\frac{x}{2}} dx \right)'$ , 得  $f'(x) = e^{\frac{x}{2}}$ , 则  $f''(x) = \left( e^{\frac{x}{2}} \right)' = e^{\frac{x}{2}} \cdot \frac{1}{2} = \frac{1}{2} e^{\frac{x}{2}}$ , 代入  $x = 0$  得

$f''(0) = \frac{1}{2} e^{\frac{0}{2}} = \frac{1}{2}$ .

3. 答案:  $\sin x + C$

解析: 原式  $= \int \frac{\cos^2 x}{\cos x} dx = \int \cos x dx = \sin x + C$ .

4. 答案:  $f(x)$

解析: 由不定积分性质可得,  $\frac{d}{dx} \left[ \int f(x) dx \right] = \left[ \int f(x) dx \right]' = f(x)$ .

5. 答案:  $\int f(x)dx$

解析: 由不定积分性质可得,  $d\left[\int f(x)dx\right] = f(x)dx$ .

6. 答案:  $-4$

解析: 由题可得,  $\int f(x)dx = 2e^{-2x} + C$ , 两边求导得  $f(x) = -4e^{-2x}$ , 则  $a = -4$ .

7. 答案:  $-\frac{1}{2}(1-x^2)^2 + C$

解析: 原式  $= -\frac{1}{2} \int f(1-x^2)(1-x^2)'dx = -\frac{1}{2} \int f(1-x^2)d(1-x^2) = -\frac{1}{2}(1-x^2)^2 + C$ .

8. 答案:  $\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$

解析: 原式  $= \int \frac{1}{(x-1)(x+1)}dx = \frac{1}{2} \int \left( \frac{1}{x-1} - \frac{1}{x+1} \right)dx = \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$ .

9. 答案:  $\arcsin e^x + C$

解析: 原式  $= \int \frac{1}{\sqrt{1-e^{2x}}}e^x dx = \int \frac{1}{\sqrt{1-e^{2x}}}(e^x)'dx = \int \frac{1}{\sqrt{1-e^{2x}}}de^x = \arcsin e^x + C$ .

10. 答案:  $x \ln x - x + C$

解析: 两边积分  $\int F'(x)dx = \int \ln x dx$  得  $F(x) = \int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C$ .

11. 答案:  $\frac{1}{5}x^5 + C$

解析: 原式  $= \int x^5 \cdot \frac{1}{x} dx = \int x^4 dx = \frac{1}{5}x^5 + C$ .

### 三、计算题

1. (1) 答案:  $\frac{2}{3}x^{\frac{3}{2}} + 2\sqrt{x} + C$

解析: 原式  $= \int \left( x^{\frac{1}{2}} + \frac{1}{\sqrt{x}} \right) dx = \frac{2}{3}x^{\frac{3}{2}} + 2\sqrt{x} + C.$

(2) 答案:  $-3x^{-\frac{1}{3}} - \frac{(e\pi)^x}{1 + \ln \pi} + C$

解析: 原式  $= \int x^{-\frac{4}{3}} dx - \int (e\pi)^x dx = -3x^{-\frac{1}{3}} - \frac{(e\pi)^x}{\ln(e\pi)} + C = -3x^{-\frac{1}{3}} - \frac{(e\pi)^x}{1 + \ln \pi} + C.$

(3) 答案:  $\frac{1}{3}x^3 + 2\ln|x| - \frac{1}{3x^3} + C$

解析: 原式  $= \int \left( x^2 + \frac{2}{x} + \frac{1}{x^4} \right) dx = \frac{1}{3}x^3 + 2\ln|x| - \frac{1}{3x^3} + C.$

(4) 答案:  $\frac{1}{2}x^2 - \ln|x| + C$

解析: 原式  $= \int \left( x - \frac{1}{x} \right) dx = \frac{1}{2}x^2 - \ln|x| + C.$

(5) 答案:  $\sin x + \ln|x| + C$

解析: 原式  $= \int \left( \cos x + \frac{1}{x} \right) dx = \sin x + \ln|x| + C.$

(6) 答案:  $e^x - 2\sqrt{x} + C$

解析: 原式  $= \int \left( e^x - \frac{1}{\sqrt{x}} \right) dx = e^x - 2\sqrt{x} + C.$

(7) 答案:  $\frac{1}{4}\sin(1+2x^2)+C$

解析: 原式  $= \frac{1}{4} \int \cos(1+2x^2)(1+2x^2)'dx = \frac{1}{4} \int \cos(1+2x^2)d(1+2x^2) = \frac{1}{4}\sin(1+2x^2)+C$ .

(8) 答案:  $\frac{1}{2}(\arcsin x)^2 + C$

解析: 原式  $= \int \arcsin x \cdot \frac{1}{\sqrt{1-x^2}} dx = \int \arcsin x \cdot (\arcsin x)' dx = \int \arcsin x d \arcsin x$

$= \frac{1}{2}(\arcsin x)^2 + C$ .

(9) 答案:  $\frac{1}{2}(\arctan x)^2 + C$

解析: 原式  $= \int \arctan x \cdot (\arctan x)' dx = \int \arctan x d \arctan x = \frac{1}{2}(\arctan x)^2 + C$ .

(10) 答案:  $-\sqrt{1-x^2} + C$

解析: 原式  $= -\frac{1}{2} \int \frac{1}{\sqrt{1-x^2}}(1-x^2)'dx = -\frac{1}{2} \int \frac{1}{\sqrt{1-x^2}}d(1-x^2) = -\sqrt{1-x^2} + C$ .

(11) 答案:  $-\frac{1}{4}\cos^4 x + C$

解析: 原式  $= -\int \cos^3 x (\cos x)' dx = -\int \cos^3 x d \cos x = -\frac{1}{4}\cos^4 x + C$ .

(12) 答案:  $\frac{1}{11}\tan^{11} x + C$

解析: 原式  $= \int \tan^{10} x \cdot (\tan x)' dx = \int \tan^{10} x d \tan x = \frac{1}{11}\tan^{11} x + C$ .

(13) 答案:  $\frac{1}{2}(x - \sin x) + C$

解析: 原式  $= \int \sin^2 \frac{x}{2} dx = \int \frac{1 - \cos(2 \cdot \frac{x}{2})}{2} dx = \frac{1}{2} \int (1 - \cos x) dx = \frac{1}{2}(x - \sin x) + C$ .

(14) 答案:  $x^3 + \arctan x + C$

解析: 原式  $= \int \frac{3x^2(x^2+1)+1}{x^2+1} dx = \int \left( 3x^2 + \frac{1}{x^2+1} \right) dx = x^3 + \arctan x + C$ .

(15) 答案:  $\tan x - x + C$

解析: 原式  $= \int (\sec^2 x - 1) dx = \int \sec^2 x dx - \int 1 dx = \tan x - x + C$ .

(16) 答案:  $2 \ln(1 + \sqrt{x}) + C$

解析: 原式  $= 2 \int \frac{1}{1+\sqrt{x}} (1+\sqrt{x})' dx = 2 \ln(1+\sqrt{x}) + C$ .

(17) 答案:  $-\cos x + \frac{1}{3} \cos^3 x + C$

解析: 原式  $= \int \sin^2 x \cdot \sin x dx = \int (1 - \cos^2 x) \sin x dx = -\int (1 - \cos^2 x) d \cos x = -\cos x + \frac{1}{3} \cos^3 x + C$ .

(18) 答案:  $-\frac{1}{x \ln x} + C$

解析: 原式  $= \int \frac{1}{(x \ln x)^2} (x \ln x)' dx = \int \frac{1}{(x \ln x)^2} d(x \ln x) = -\frac{1}{x \ln x} + C$ .

(19) 答案:  $e^{\tan x} + C$

解析: 原式  $= \int e^{\tan x} (\tan x)' dx = \int e^{\tan x} d \tan x = e^{\tan x} + C$ .

(20) 答案:  $-\frac{1}{2} (\ln \cos x)^2 + C$

解析: 原式  $= -\int \ln \cos x (\ln \cos x)' dx = -\int \ln \cos x d \ln \cos x = -\frac{1}{2} (\ln \cos x)^2 + C$ .



(21) 答案:  $\frac{1}{10}(2x+1)^{\frac{5}{2}} - \frac{1}{6}(2x+1)^{\frac{3}{2}} + C$

解析: 令  $\sqrt{2x+1} = t$ , 则  $x = \frac{t^2-1}{2}$ ,  $dx = tdt$ ,

原式  $= \int \frac{t^2-1}{2} \cdot t \cdot tdt = \frac{1}{2} \int (t^4 - t^2) dt = \frac{1}{10}t^5 - \frac{1}{6}t^3 + C = \frac{1}{10}(2x+1)^{\frac{5}{2}} - \frac{1}{6}(2x+1)^{\frac{3}{2}} + C$ .

(22) 答案:  $\frac{1}{6}(2x-1)^{\frac{3}{2}} - \frac{1}{4}(2x-1) + C$

解析: 令  $\sqrt{2x-1} = t$ , 则  $x = \frac{t^2+1}{2}$ ,  $dx = tdt$ ,

原式  $= \int \frac{\frac{t^2+1}{2}-1}{1+t} \cdot tdt = \frac{1}{2} \int (t-1)tdt = \frac{1}{2} \int (t^2 - t)dt = \frac{1}{6}t^3 - \frac{1}{4}t^2 + C = \frac{1}{6}(2x-1)^{\frac{3}{2}} - \frac{1}{4}(2x-1) + C$ .

(23) 答案:  $-\frac{2}{3}(1-x)^{\frac{3}{2}} + \frac{4}{5}(1-x)^{\frac{5}{2}} - \frac{2}{7}(1-x)^{\frac{7}{2}} + C$

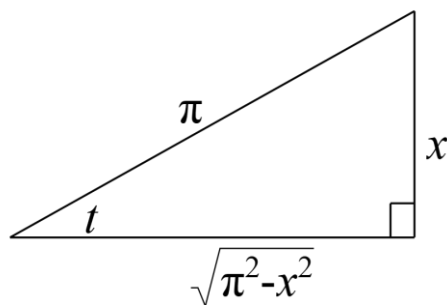
解析: 令  $\sqrt{1-x} = t$ , 则  $x = 1-t^2$ ,  $dx = -2tdt$ ,

原式  $= \int (1-t^2)^2 \cdot t \cdot (-2t)dt = -2 \int (t^2 - 2t^4 + t^6)dt = -2 \left( \frac{1}{3}t^3 - \frac{2}{5}t^5 + \frac{1}{7}t^7 \right) + C$

$= -\frac{2}{3}(1-x)^{\frac{3}{2}} + \frac{4}{5}(1-x)^{\frac{5}{2}} - \frac{2}{7}(1-x)^{\frac{7}{2}} + C$ .

(24) 答案:  $\frac{\pi^2}{2} \arcsin \frac{x}{\pi} + \frac{x}{2} \sqrt{\pi^2 - x^2} + C$

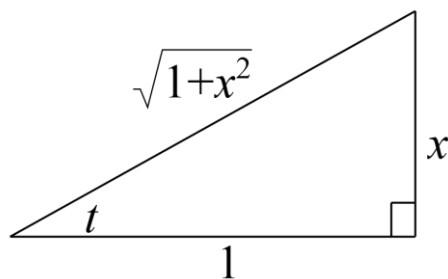
解析：令  $x = \pi \sin t$ ， $dx = \pi \cos t dt$ ，



$$\text{原式} = \int \pi \cos t \cdot \pi \cos t dt = \pi^2 \int \frac{1 + \cos 2t}{2} dt = \frac{\pi^2}{2} t + \frac{\pi^2}{4} \sin 2t + C = \frac{\pi^2}{2} \arcsin \frac{x}{\pi} + \frac{x}{2} \sqrt{\pi^2 - x^2} + C.$$

(25) 答案：  $\ln(x + \sqrt{x^2 + 1}) + C$

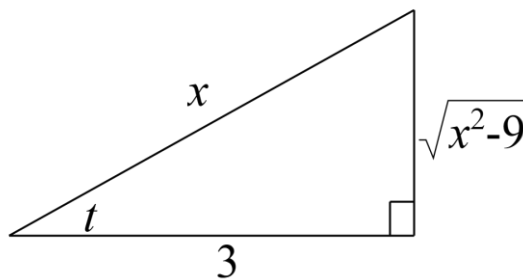
解析：令  $x = \tan t$ ，则  $dx = \sec^2 t dt$ ，



$$\text{原式} = \int \frac{1}{\sec t} \cdot \sec^2 t dt = \int \sec t dt = \ln |\sec t + \tan t| + C = \ln(x + \sqrt{x^2 + 1}) + C.$$

(26) 答案：  $\sqrt{x^2 - 9} - 3 \arccos \frac{3}{x} + C$

解析：令  $x = 3 \sec t$ ，则  $dx = 3 \sec t \tan t dt$ ，

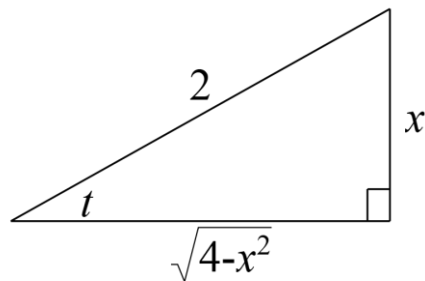


$$\text{原式} = \int \frac{3 \tan t}{3 \sec t} \cdot 3 \sec t \tan t dt = 3 \int \tan^2 t dt = 3 \int (\sec^2 t - 1) dt = 3(\tan t - t) + C$$

$$= \sqrt{x^2 - 9} - 3 \arccos \frac{3}{x} + C.$$

(27) 答案:  $\frac{x}{4\sqrt{4-x^2}} + C$

解析: 令  $x = 2 \sin t$ , 则  $dx = 2 \cos t dt$ ,



$$\text{原式} = \int \frac{1}{(2 \cos t)^3} \cdot 2 \cos t dt = \frac{1}{4} \int \sec^2 t dt = \frac{1}{4} \tan t + C = \frac{x}{4\sqrt{4-x^2}} + C.$$

(28) 答案:  $x \sin x + \cos x + C$

解析: 原式  $= \int x \sin x = x \sin x - \int 1 \cdot \sin x dx = x \sin x + \cos x + C.$

(29) 答案:  $\frac{1}{2} x^2 \ln 3x - \frac{1}{4} x^2 + C$

解析: 原式  $= \int \ln 3x d\frac{1}{2}x^2 = \frac{1}{2}x^2 \ln 3x - \int \frac{1}{x} \cdot \frac{1}{2}x^2 dx = \frac{1}{2}x^2 \ln 3x - \frac{1}{2} \int x dx = \frac{1}{2}x^2 \ln 3x - \frac{1}{4}x^2 + C.$

(30) 答案:  $xe^x - e^x + C$

解析: 原式  $= \int x de^x = xe^x - \int 1 \cdot e^x dx = xe^x - e^x + C.$

(31) 答案:  $-\frac{1}{2}x \cos 2x + \frac{1}{4} \sin 2x + C$

解析: 原式  $= \int x d\left(-\frac{1}{2} \cos 2x\right) = -\frac{1}{2}x \cos 2x + \frac{1}{2} \int \cos 2x dx = -\frac{1}{2}x \cos 2x + \frac{1}{4} \sin 2x + C.$

(32) 答案:  $-\frac{1}{4}x \cos 2x + \frac{1}{8} \sin 2x + C$

解析: 原式  $= \int x \cdot \frac{1}{2} \sin 2x dx = \frac{1}{2} \int x d\left(-\frac{1}{2}x \cos 2x\right) = -\frac{1}{4}x \cos 2x + \frac{1}{4} \int \cos 2x dx$   
 $= -\frac{1}{4}x \cos 2x + \frac{1}{8} \sin 2x + C.$

(33) 答案:  $(x+3)\ln(x+3) - x + C$

解析: 原式  $= (x+3)\ln(x+3) - \int \frac{1}{x+3} \cdot (x+3) dx = (x+3)\ln(x+3) - x + C.$

(34) 答案:  $x \tan x + \ln|\cos x| + C$

解析: 原式  $= x \tan x - \int 1 \cdot \tan x dx = x \tan x + \ln|\cos x| + C.$

(35) 答案:  $x \arctan \frac{1}{x} + \frac{1}{2} \ln(1+x^2) + C$

解析: 原式  $= x \arctan \frac{1}{x} - \int \frac{1}{1+\frac{1}{x^2}} \left(-\frac{1}{x^2}\right) \cdot x dx = x \arctan \frac{1}{x} - \int \frac{1}{1+x^2} x dx$   
 $= x \arctan \frac{1}{x} + \frac{1}{2} \int \frac{1}{1+x^2} d(1+x^2) = x \arctan \frac{1}{x} + \frac{1}{2} \ln(1+x^2) + C.$

(36) 答案:  $2\sqrt{1+e^x} + \ln \frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C$

解析: 令  $\sqrt{1+e^x} = t$ , 则  $x = \ln(t^2-1)$ ,  $dx = \frac{2t}{t^2-1} dt$ ,

原式  $= \int t \cdot \frac{2t}{t^2-1} dt = 2 \int \frac{t^2-1+1}{t^2-1} dt = 2 \int \left(1 + \frac{1}{t^2-1}\right) dt = 2t + \ln \left| \frac{t-1}{t+1} \right| + C$

$= 2\sqrt{1+e^x} + \ln \frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C.$

(37) 答案:  $\arctan(x+2) + C$

解析：原式  $= \int \frac{1}{(x+2)^2+1} d(x+2) = \arctan(x+2) + C$  .

(38) 答案：  $\frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + C$

解析：原式  $= \frac{1}{4} \int \left( \frac{1}{x-2} - \frac{1}{x+2} \right) dx = \frac{1}{4} \ln|x-2| - \frac{1}{4} \ln|x+2| + C = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + C$  .

(39) 答案：  $\frac{1}{2} x^2 - \frac{1}{2} \ln(1+x^2) + C$

解析：原式  $= \int \frac{x^3+x-x}{1+x^2} dx = \int \frac{x(x^2+1)-x}{1+x^2} dx = \int \left( x - \frac{x}{1+x^2} \right) dx = \frac{1}{2} x^2 - \frac{1}{2} \int \frac{1}{1+x^2} (1+x^2)' dx$   
 $= \frac{1}{2} x^2 - \frac{1}{2} \ln(1+x^2) + C$  .

(40) 答案：  $\ln(1+\sin^2 x) + C$

解析：原式  $= \int \frac{1}{1+\sin^2 x} (1+\sin^2 x)' dx = \int \frac{1}{1+\sin^2 x} d(1+\sin^2 x) = \ln(1+\sin^2 x) + C$  .

(41) 答案：  $\frac{3}{2} \sqrt[3]{(x+2)^2} - 3\sqrt[3]{x+2} - 3 \ln \left| 1 + \sqrt[3]{x+2} \right| + C$

解析：令  $\sqrt[3]{x+2} = t$ ,  $x = t^3 - 2$ ,  $dx = 3t^2 dt$ ,

原式  $= \int \frac{3t^2}{1+t} dt = 3 \int \frac{t^2-1+1}{1+t} dt = 3 \int \left( t-1 + \frac{1}{1+t} \right) dt = \frac{3}{2} t^2 - 3t + 3 \ln(1+t) + C$   
 $= \frac{3}{2} \sqrt[3]{(x+2)^2} - 3\sqrt[3]{x+2} - 3 \ln \left| 1 + \sqrt[3]{x+2} \right| + C$  .

(42) 答案：  $2\sqrt{x} - 4\sqrt[4]{x} + 4 \ln(1+\sqrt[4]{x}) + C$

解析：令  $t = \sqrt[4]{x}$ ,  $x = t^4$ ,  $dx = 4t^3 dt$ ,

$$\text{原式} = \int \frac{4t^3}{t^2+t} dt = 4 \int \left( t - 1 + \frac{1}{1+t} \right) dt = 2t^2 - 4t + 4 \ln(1+t) + C = 2\sqrt{x} - 4\sqrt[4]{x} + 4 \ln(1 + \sqrt[4]{x}) + C.$$

(43) 答案:  $(x+1)\arctan \sqrt{x} - \sqrt{x} + C$

解析: 令  $t = \sqrt{x}$ ,  $x = t^2$ ,  $dx = 2t dt$ ,

$$\begin{aligned} \text{原式} &= \int \arctan t \cdot 2t dt = \int \arctan t dt^2 = t^2 \arctan t - \int \frac{t^2}{1+t^2} dt = t^2 \arctan t - \int \frac{t^2+1-1}{1+t^2} dt \\ &= t^2 \arctan t - \int \left( 1 - \frac{1}{1+t^2} \right) dt = t^2 \arctan t - t + \arctan t + C = (x+1)\arctan \sqrt{x} - \sqrt{x} + C. \end{aligned}$$

(44) 答案:  $x - 2 \ln(\sqrt{1+e^x} + 1) + C$

解析: 令  $\sqrt{1+e^x} = t$ , 则  $x = \ln(t^2 - 1)$ ,  $dx = \frac{2t}{t^2 - 1} dt$ ,

$$\begin{aligned} \text{原式} &= \int \frac{1}{t} \cdot \frac{2t}{t^2 - 1} dt = \int \left( \frac{1}{t-1} - \frac{1}{t+1} \right) dt = \ln|t-1| - \ln|t+1| + C = \ln \left| \frac{t-1}{t+1} \right| + C \\ &= \ln \left| \frac{\sqrt{1+e^x} - 1}{\sqrt{1+e^x} + 1} \right| + C = x - 2 \ln(\sqrt{1+e^x} + 1) + C. \end{aligned}$$

(45) 答案:  $(2x-4)\sqrt{e^x-1} + 4\arctan \sqrt{e^x-1} + C$

$$\begin{aligned} \text{解析: 令 } \sqrt{e^x-1} = t, \text{ 则 } x = \ln(t^2+1), \quad dx &= \frac{2t}{t^2+1} dt, \quad \text{原式} = \int \frac{\ln(t^2+1) \cdot e^{\ln(t^2+1)}}{t} \cdot \frac{2t}{t^2+1} dt \\ &= 2 \int \ln(t^2+1) dt = 2t \ln(t^2+1) - 2 \int \frac{2t}{t^2+1} \cdot t dt = 2t \ln(t^2+1) - 4 \int \frac{t^2+1-1}{t^2+1} dt = 2t \ln(t^2+1) - 4 \int \left( 1 - \frac{1}{t^2+1} \right) dt \\ &= 2t \ln(t^2+1) - 4t + 4 \arctan t + C = (2x-4)\sqrt{e^x-1} + 4 \arctan \sqrt{e^x-1} + C. \end{aligned}$$

(46) 答案:  $-\frac{1}{2}e^{-x^2} + x \ln x - x + C$

解析：原式  $= \int x e^{-x^2} dx + \int \ln x dx = -\frac{1}{2} \int e^{-x^2} (-x^2)' dx + x \ln x - \int \frac{1}{x} \cdot x dx$   
 $= -\frac{1}{2} \int e^{-x^2} d(-x^2) + x \ln x - x + C = -\frac{1}{2} e^{-x^2} + x \ln x - x + C.$

2. 答案：  $-\frac{1}{3}(1-x^2)^{\frac{3}{2}} + C$

解析：两边求导  $\left[ \int x f(x) dx \right]' = (\arcsin x + C)'$  得  $x f(x) = \frac{1}{\sqrt{1-x^2}}$ ，则  $\frac{1}{f(x)} = x \sqrt{1-x^2}$ ，

原式  $= \int x \sqrt{1-x^2} dx = -\frac{1}{2} \int \sqrt{1-x^2} (1-x^2)' dx = -\frac{1}{2} \int \sqrt{1-x^2} d(1-x^2) = -\frac{1}{3} (1-x^2)^{\frac{3}{2}} + C.$

3. 答案：  $\frac{1}{2} \sin^2 x^2 - \frac{1}{x} + C$

解析：两边求导  $\left( \int f(x) dx \right)' = (\sin^2 x + C)'$ ，得  $f(x) = \sin 2x$ ，原式

$= \int \frac{x^2 f(x^2)}{x} dx + \int \frac{1}{x^2} dx = \int x f(x^2) dx - \frac{1}{x} = \frac{1}{2} \int f(x^2) dx^2 - \frac{1}{x} = \frac{1}{2} \sin^2 x^2 - \frac{1}{x} + C.$

4. 答案：(1)  $f(x) = x^3 - x^2 + 2$ ；(2)  $\frac{1}{4} x^4 + \frac{5}{3} x^3 - x^2 + 2x + C$

解析：由题意  $f'(x) = 3x^2 - 2x$ ，两边积分  $\int f'(x) dx = \int (3x^2 - 2x) dx$ ，得  $f(x) = x^3 - x^2 + C$ ，将

$f(1) = 2$  代入得  $1 - 1 + C = 2$ ，解得  $C = 2$ ，故  $f(x) = x^3 - x^2 + 2$ ， $f'(x) = 3x^2 - 2x$ ，

$f''(x) = 6x - 2$ ，故  $\int [f(x) + x f''(x)] dx = \int (x^3 - x^2 + 2 + 6x^2 - 2x) dx = \int (x^3 + 5x^2 - 2x + 2) dx$

$\int (x^3 + 5x^2 - 2x + 2) dx = \frac{1}{4} x^4 + \frac{5}{3} x^3 - x^2 + 2x + C.$

#### 四、证明题

1. 答案：见解析

证明：令  $x = a \sin t$ ， $dx = a \cos t dt$ ，原式  $= \int \frac{1}{a \cos t} \cdot a \cos t dt = \int 1 dt = t + C$ ，

由  $x = a \sin t$  得  $t = \arcsin \frac{x}{a}$ , 所以  $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \frac{x}{a} + C (a > 0)$ , 得证.